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UNSUPERVISED LEARNING OF DISENTANGLED SPEECH CONTENT AND STYLE REPRESENTATION

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ABSTRACT

We present an approach for unsupervised learning of speech representation disentangling contents and styles. Our model consists of: (1) a local encoder that captures per-frame information; (2) a global encoder that captures per-utterance information; and (3) a conditional decoder that reconstructs speech given local and global latent variables. Our experiments show that (1) the local latent variables encode speech contents, as reconstructed speech can be recognized by ASR with low word error rates (WER), even with a different global encoding; (2) the global latent variables encode speaker style, as reconstructed speech shares speaker identity with the source utterance of the global encoding. Additionally, we demonstrate an useful application from our pre-trained model, where we can train a speaker recognition model from the global latent variables and achieve high accuracy by fine-tuning with as few data as one label per speaker.

***Index Terms*—** Unsupervised, Disentanglement, Speech, VQVAE

1. INTRODUCTION

Speech waveforms are a complex, high-dimensional form of data influenced by a number of underlying factors, which can be broadly categorized into linguistic contents and speaking styles. Learning disentangled latent representations from speech has a wide set of applications in generative tasks, including speech synthesis, data augmentation, voice transfer, and speech compression. Downstream tasks such as speech recognition [1] and speaker classification [2] can also benefit from such learned representations.

Because of the cost, complexity, and privacy concerns around collecting labeled speech data, there has been a lot of interest in unsupervised representation learning for speech. Of particular interest is to learn representations for speech styles from unsupervised data due to the difficulty in describing prosody with human labels.

Some previous works aim to learn *global* representations from entire speech sequences. Global style tokens [3] learn a dictionary of embeddings from speech without prosody labels. As another example, Hsu et al. [4] model disentangled speech styles with a hierarchy of variational autoencoder (VAE) [5]. Hu et al. [6] proposed a content and style separation model by pre-training on a single-speaker dataset with text transcription and minimizing mutual information (MI) between the content and style representation.

Other works try to learn fine-grain *localized* representations of speech. [2, 7, 8] apply self-supervised learning to unlabeled speech data and extract localized latent representations that can be fine-tuned for speech recognition. FHVAE [9] learns a sequence of high-level features by applying VAE to every frame. [10, 11, 12] leverages vector-quantized VAE [13] (VQVAE) to learn a discrete sequence representation of speech.

We propose a framework to learn both global and localized representation of speech. In order to disentangle content and style representations, we apply (1) a local encoder with VQ layer to learn a discrete per-timestep representation of the speech that captures the linguistic contents and (2) a global VAE to extraction per-utterance representations to reflect the speech styles. We further disentangle the local and global representations with a mutual information loss [14]. We evaluate the quality of linguistic and style representations by running speech and speaker recognition models on the reconstructed speech. We also show that the global representation captures the speaker information well enough that we can obtain a speaker classification model by training a linear projection layer on top of the global representation with only one example per speaker.

2. MODEL ARCHITECTURE

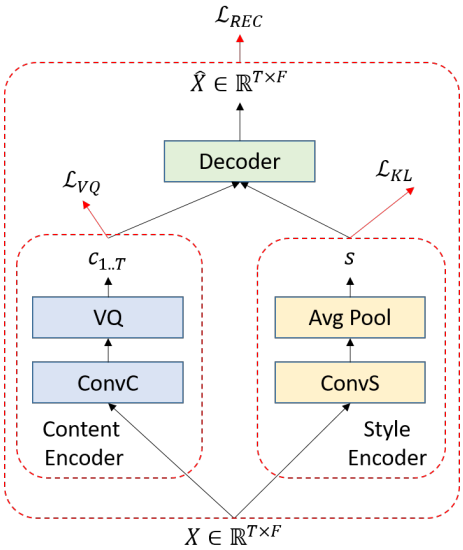
Our proposed model is illustrated in Figure 1. On the encoder side, we have two different encoders with different architectures, encoding content and style respectively. On the decoder side, we have a decoder that takes both content and style latent variables to produce speech features.

2.1. Content Encoder

The content encoder $EncC(\cdot)$ aims to extract a sequence of latent variables $C = [c_1, .., c_T] \in \mathbb{R}^{T \times D_C}$ that only repre-

*This work was done while the first author was a research intern at Google Brain.

Fig. 1. Proposed model: disentangling content and style autoencoder.



sent the content from the input speech $X \in \mathbb{R}^{T\times F}$. Inside the content encoder, we have 10 1D convolutional layers with residual connections. We use time stride 2 in the 3rd layer, thus final output length was reduced from T to $T/2$.

On the top of the content encoder, we apply information bottleneck via vector quantization (VQ) [13] with straight-through gradients. This information bottleneck forces the content encoder to capture only necessary localized information such as phoneme or subword-like representation and discard the speaking style variation. Without using information bottleneck or strong regularization method such as VQ layer, the content encoder might capture both content and style inside $c_{1..T}$ and the decoder will ignore the style latent variable s from the style encoder.

2.2. Style Encoder

The style encoder $EncS(\cdot)$ aims to extract a global latent variable $s \in \mathbb{R}^{D_S}$ that only represent the speaking style from the input speech X. The input X is processed by 6 residual 1D convolution layers with time stride 2 on three different layers, resulted in 8× time-length reduction.

We summarize the output from the convolutional layer into a single vector with a global average pooling operation across the time-axis. We put a variational layer with Gaussian posterior [5] on the top of convolutional layer output. During training, the style variable s is sampled from Eq. 3. In the inference step, we directly use s_μ to represent the style information.

2.3. Decoder

The decoder $Dec(.,.)$ takes both content and style latent variables $[c_{1..T}, s]$ and reconstruct the features $\hat{X} \in \mathbb{R}^{T\times F}$ during training stage. For the inference step, we can combine $c_{1..T}$ and s from different input speech and generate a new speech feature. We use 10 1D convolution layers with residual connection and we feed the style information by concatenating s in the channel axis on {1,3,5,7}-th layers.

2.4. Loss

We optimize our model with multiple loss functions:

1. **Reconstruction loss**: $\mathcal{L}_{REC} = \|X - \hat{X}\|_1^2 + \|X - \hat{X}\|_2^2$. The reconstruction loss minimizes both L1 and L2-norm squared distance between groundtruth X and $\hat{X}$.

2. **Content VQ loss**:

$$i = argmin_j \|ConvC(X) - E_j\|_2 \quad (1)$$

$$\mathcal{L}_{VQ} = \|ConvC(X) - sg(E_i)\|_2^2 \quad (2)$$

where $sg(\cdot)$ is stop gradient operations. The content VQ loss encourages the encoder output z to minimize the distance between itself and the nearest codebook E_i.

3. **Style regularization loss**:

$$s_\mu, s_\sigma^2 = EncS(X); \quad s \sim \mathcal{N}(s_\mu, s_\sigma^2) \quad (3)$$

$$\mathcal{L}_{KL} = \mathbb{D}_{KL}(\mathcal{N}(s_\mu, s_\sigma^2) \| \mathcal{N}(0,1)) \quad (4)$$

The style regularization loss minimizes the KL divergence between the Gaussian posterior $\mathcal{N}(s_\mu, s_\sigma^2)$ with the unit Gaussian prior $\mathcal{N}(0,1)$.

The final loss is a combined sum: $Ł = Ł_{REC} + 0.25Ł_{VQ} + Ł_{KL}$.

2.5. Mutual Information Estimation and Minimization

Section 2.1 describes the importance of information bottleneck via VQ layer on the content encoder sides. However, the quantized variables $c_{1..T}$ may still manage to carry non-content information such as speaking style.

To further decouple content and style representations without explicit labels, we estimate and minimize the mutual information (MI) from the output between content $c_{1..T}$ with style s. By minimizing the mutual information, we reduce the amount of correlation between two random variables [15].

Measuring mutual information directly from high dimensional random variables is a difficult problem [16]. Here, we use InfoNCE [2] to estimate the lower bound MI between content and style:

$$C_i = AvgPool(ConvC(X_i)) \quad (5)$$
$$S_i = AvgPool(ConvS(X_i))$$

$$I(X;Y) \geq \mathbb{E}\left[\frac{1}{K}\sum_{i=1}^{K} \log \frac{e^{Sc(C_i,S_i)}}{\frac{1}{K}\sum_{j=1}^{K} e^{Sc(C_i,S_j)}}\right] \triangleq I_{NCE}.$$

where K is the number of samples (across different utterances), $Sc(\cdot,\cdot)$ is a neural network scorer, C_i is the representation from content encoder prior to the quantization, and S_i is the representation from style encoder.

Finally, we simultaneously trying to maximize I_{NCE} w.r.t Sc and minimize I_{NCE} and Ł w.r.t ($\theta = \{EncC, EncS, Dec\}$):

$$\max_{Sc} \min_{EncC,EncS,Dec} (Ł + I_{NCE}). \quad (6)$$

We apply adaptive gradient scaling [14] to stabilize the loss Eq. 6 during training. For each step, the model parameters are updated with:

$$g_\theta = \partial Ł/\partial\theta; \quad g_a = \partial I_{NCE}/\partial\theta \quad (7)$$

$$g_b = min(\|g_a\|_2, \|g_\theta\|_2)\frac{g_a}{\|g_a\|_2} \quad (8)$$

$$\theta = \theta - (g_\theta + g_b) \quad (9)$$

$$Sc = Sc + \partial I_{NCE}/\partial Sc \quad (10)$$

3. EXPERIMENTS

3.1. Dataset

To evaluate our proposed model effectiveness on disentangling content and style, we use the Librispeech [17] dataset for all following experiments. To train the generative model, we use all training sets: train-clean 100 hours, train-clean 360 hours and train-other 500 hours. We use dev-clean as the development set and test-clean for evaluation. In total, Librispeech contains 2484 unique speakers. For the feature extraction, we extract 80 dimensions log Mel-filterbank with window size 25 ms and step size 10 ms. All features are normalized into zero mean and unit standard deviation.

3.2. Hyperparameters.

We evaluate variations of the model along two dimensions. First, we measure the model performance with different VQ-VAE codebook sizes with $K \in \{256, 512, 1024\}$. Second, we compare the disentanglement performance between models trained with vs. without the mutual information minimization loss. All models are optimized with Adam[18] optimizer up to 800,000 steps. In total, our model has $\pm$30 million parameters.

3.3. Experiments on Content Preservation

This set of experiments aim to measure how well the content encoder extracts and preserves the original content from the input speech. There are two different scenarios:

1. No shuffle: feeds speech X_i into both content and style encoder and predict the speech features $\hat{X}$ with the decoder. Since each frame is encoded by 8 to 10 bits of VQVAE codes, this setup tests how well the model compresses speech.

Table 1. Predicted features WER between different codebook sizes and with or without mutual information (MI) loss. The Conformer L model's WER on the original test-clean set is 2.1%.

Codebook Size	MI Loss	WER (%) No shuffle	WER (%) Shuffle
256	No	4.6	7.7
	Yes	4.4	7.0
512	No	4.2	7.0
	Yes	3.8	6.8
1024	No	3.7	6.3
	Yes	3.6	6.1

2. Shuffle: feeds speech X_i into the content encoder and speech X_j ($i \neq j$) into the style encoder and predict the speech features $\hat{X}$ with the decoder. This setup measures whether the speech contents are preserved by the VQVAE codes.

We use test-clean for both content and style encoder in order to test the performance of our system with unseen content and speaking style. To evaluate how similar the predicted speech content compared to the original content from speech X_i, we utilize a state-of-the-art pre-trained Conformer L [19] to transcribe the predicted features $\hat{X}$. After that, we calculate the word error rate for the transcription result with the ground-truth text from input speech X_i on the content encoder side.

Based on Table 1, we observed that increasing the codebook size allows the model to better preserve speech contents, as manifested by the decreasing WER for both shuffle and no shuffle scenarios. Adding the mutual information loss further improves content preservation with decreased WER across all code book sizes, for both shuffle and no-shuffle scenarios.

3.4. Experiments on Style Disentanglement

This set of experiments measure how well the style encoder extracts and preserves the original speaking style from the input speech. We feed different input speech X_i into the content encoder and input speech X_j ($i \neq j$) into the style encoder and predict speech features $\hat{X}$ based on the content and style latent variables.

Next, we evaluate the speaker similarity between the generated speech and X_i and X_j, the input speeches for the content and style encoders, respectively:

1. Train a speaker recognition model using ground-truth speaker labels;

2. Predict the softmax output on the predicted features $\hat{X}$.

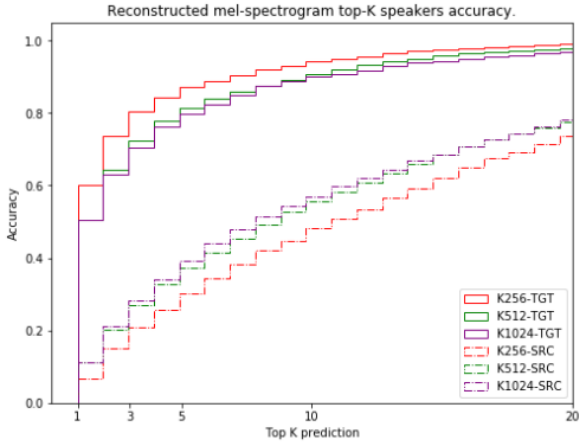
Based on the softmax output predictions, we calculate several metrics such as: average ranking, top-1, top-3, and top-5 accuracy compared to the ground truth speaker on X_j. As shown in Figure 2, the generated speech has a speaker style

Table 2. Predicted features top-K speaker ranks between different K codebook size and with or without mutual information loss.

Codebook	MI	Avg.	Top-1	Top-3	Top-5
256	No	3.9	0.5	0.72	0.81
	Yes	2.9	0.6	0.8	0.87
512	No	3.9	0.5	0.71	0.8
	Yes	3.7	0.5	0.73	0.81
1024	No	4.0	0.52	0.72	0.8
	Yes	4.0	0.51	0.71	0.8

much closer to X_j, the input to the style encoder, than X_i, the input to the content encoder. Table 2 further shows how the speaker style accuracy for X_j changes as we vary codebook sizes and whether to enable mutual information loss. We can see that the style and content information becomes more entangled as we increase the codebook size. The mutual information loss helps improving the disentanglement, esp. for the smaller codebooks. Table 1 and Table 2 together highlight the tradeoffs between content reconstruction and style disentanglement.

Fig. 2. Top-k speaker accuracy compared to content (SRC) speaker ID and style (TGT) speaker ID. In this figure, we varied the codebook size and all models are trained with MI loss.



3.5. Experiments on Few-Shot Speaker Recognition

Recently, there are many success stories for pre-training a model with a large amount of unlabeled data and fine-tune it with a small labeled data [1, 20]. In this experiment, we fine-tune a pre-trained style encoder for the few-shot speaker recognition task. As shown in Figure 3, we add a linear projection layer on the top of the style encoder to compute the logits for speaker classification. During the fine-tuning, we freeze all parameters except the final softmax projection layer.

We consider three scenarios for fine-tuning a speaker recognition model: 1) train from scratch without pre-trained weight (without freezing any weight), 2) initialize with a style encoder pre-trained without MI loss (freeze style encoder weight), 3) initialize with a style encoder pre-trained with MI loss (freeze style encoder weight). We also vary the amount of fine-tuning data with 1 or 3 examples per speaker, termed as "1-shot" and "3-shot", respectively, in Table 3. The results show that we can achieve $> 90\%$ accuracy with 1-shot fine-tuning, compared with $< 11\%$ without a pre-trained encoder, and $> 95\%$ accuracy with 3 examples per speaker, compared with 63% without pre-training. Further, the use of mutual information loss brings small improvements to the results.

Fig. 3. Using pre-trained style encoder as a speaker recognition by appending and fine-tuning a softmax layer.

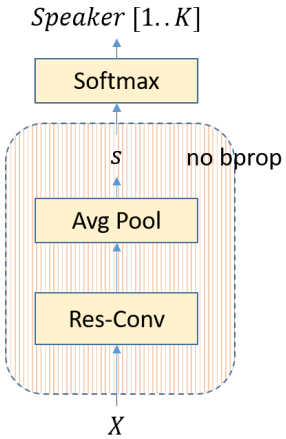


Table 3. 1-shot and 3-shot accuracy based on three difference scenarios.

Model	1-shot acc (%)	3-shot acc (%)
M1	10.8	63.0
M2	90.5	95.9
M3	91.4	96.4

4. CONCLUSION

We present a framework to learn disentangled representations of speech with unlabeled data. The framework includes a local VQ encoder to extract a discrete sequence representation of the speech contents and a global VAE encoder to learn a continuous representation of speech styles. Our evaluation shows that the discrete sequence representation effectively captures the linguistic contents while the continuous global representation encapsulates the speaker style. Additionally, we also show the application from our pre-trained model, in which we successfully train a speaker recognition system with high accuracy ($> 90\%$) only with one sample per speaker.

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